

Problem 1

A quartz nugget with gold has a mass of $m = 51$ g and a density of $\rho = 8500$ kg/m³. How much gold is in this nugget?

Solution

Let the masses and densities of gold and quartz be:

$$m = m_1 + m_2, \quad \frac{m}{V} = \rho.$$

Where: - m_1 and ρ_1 are the mass and density of quartz (2650 kg/m³). - m_2 and ρ_2 are the mass and density of gold (19300 kg/m³).

The volume of the nugget is:

$$V = \frac{m}{\rho} = \frac{0.051}{8500} = 6 \times 10^{-6} \text{ m}^3.$$

For gold and quartz, we have:

$$V = V_1 + V_2, \quad V_1 = \frac{m_1}{\rho_1}, \quad V_2 = \frac{m_2}{\rho_2}.$$

Substitute:

$$\frac{m_1}{2650} + \frac{m_2}{19300} = 6 \times 10^{-6}.$$

Since $m_1 + m_2 = 51$ g:

$$m_1 = 51 \text{ g} - m_2.$$

Substitute and solve:

$$\frac{51 - m_2}{2650} + \frac{m_2}{19300} = 6 \times 10^{-6}.$$

Multiply by the least common multiple:

$$(51 - m_2) \frac{19300}{2650} + m_2 = 116.$$

Expand:

$$(51 - m_2)(7.28) + m_2 = 116.$$

$$371.28 - 7.28m_2 + m_2 = 116.$$

Combine like terms:

$$371.28 - 6.28m_2 = 116.$$

Solve for m_2 :

$$m_2 = \frac{371.28 - 116}{6.28} \approx 40 \text{ g}.$$

Thus, the mass of gold in the nugget is:

$$m_2 = 40 \text{ g}.$$